

PRD_Inventory_Module

Enterprise Inventory, Warehouse and Stock Control

Full Product Requirements Document for Product, UX, Frontend, Backend, QA, Security and DevOps Teams

Implementation principle: build the complete inventory capability first. All pages, sections, widgets, workflows and actions must be compatible with Platform Configuration. User-type restrictions and external integrations are added later.

Status: Confirmed product direction

Version: 1.0

Priority: Core ERP module

Audience: Full-stack development team

1. Document Control

Field	Value
Document	PRD_Inventory_Module
Primary pages	Inventory Dashboard, Products, Product Page, Warehouses, Warehouse Page, Stock, Movements, Counts, Valuation, Reports, Settings
Connected modules	Group, Company, Departments, Teams, Projects, Assets, Procurement, Finance, Documents, Tasks, Contracts, Users
Initial access model	Full capability first; role restrictions later
Platform Configuration	Mandatory at module, page, section, widget, feature, workflow and action level
Source-of-truth rule	Inventory owns stock quantity, location, movement and valuation context; Finance owns accounting postings; Procurement owns purchasing workflows

2. Executive Summary

The Inventory Module is the platform-wide source of truth for products, materials, consumables, spare parts, finished goods and stock held across companies, warehouses, project sites and internal locations. It supports receiving, issuing, transferring, reserving, counting, tracking, costing and reporting without duplicating Procurement, Finance, Assets or Projects records.

- Universal architecture for construction, real estate, logistics, manufacturing, retail, hospitality, healthcare and services.
- Multi-company and multi-warehouse from the first release.
- Support for serial, batch, lot and expiration tracking.
- Support for product units, conversions, kits and bills of materials.
- Complete auditability of every stock movement.
- Operational and financial valuation views.
- Full responsive behavior and premium interaction design.

Inventory is not a spreadsheet of quantities. It is a controlled stock ledger with immutable movements and reproducible balances.

3. Product Goals

- Know exactly what is in stock, where it is, who controls it and what it is worth.
- Prevent negative, duplicate or untraceable stock movements.
- Connect purchasing, receiving, storage, project consumption, maintenance and finance.
- Support both simple and advanced warehouse operations.
- Provide fast search, scanning and mobile workflows.
- Enable configuration by business type without changing the core data model.
- Scale from one store room to a multi-company warehouse network.

4. Non-Goals

- Inventory does not replace Procurement approval and supplier selection.
- Inventory does not own accounting journals or statutory financial statements.
- Inventory does not replace Assets for long-lived assigned equipment.
- Initial release excludes live courier, IoT, RFID-reader and marketplace integrations.
- Initial release does not define final user permissions.

5. Inventory Classification Model

- Product / resale item
- Raw material
- Construction material

- Consumable
- Spare part
- Finished good
- Work in progress
- Packaging
- Tool inventory
- Rental item
- Service item (non-stock)
- Kit / bundle
- Custom type

Inventory Item Type, Product Category and Accounting Category are separate concepts and must not be conflated.

6. Information Architecture

- Inventory Dashboard
- Products Module
- Product Page
- Categories
- Brands
- Units of Measure
- Warehouses
- Warehouse Page
- Locations, Zones, Aisles, Shelves and Bins
- Stock
- Stock Movements
- Transfers
- Reservations
- Purchase Receipts
- Goods Issues
- Returns
- Cycle Counts
- Physical Inventory
- Serial Numbers
- Batch and Lot Tracking
- Expiration Management
- Kits and Bundles
- Bills of Materials
- Reorder Rules
- Supplier Inventory
- Valuation and Costing
- Analytics
- Reports
- Inventory Settings

7. Inventory Dashboard

- Total inventory value
- Total stock quantity
- Available stock
- Reserved stock
- In-transit stock
- Low-stock items
- Out-of-stock items
- Overstock items
- Expiring items
- Open receipts
- Open issues

- Pending transfers
- Count variances
- Dead stock
- Fast-moving items
- Recent movements
- Warehouse comparison

Every KPI must drill down to the underlying products, locations and movements.

8. Inventory Module Header

- Add Product
- Create Receipt
- Create Issue
- Create Transfer
- Start Count
- Global Search
- Advanced Filters
- Saved Views
- Import
- Export
- Scanner action
- List/Grid/Table switch

9. Products Module

- Product cards and rows
- Bulk create/import
- Bulk category assignment
- Bulk status update
- Duplicate detection
- Archive and restore
- Saved views
- Column chooser
- Quick stock preview
- Quick action menu

10. Product Card and Row

- Product image
- Product name
- SKU
- Barcode
- Category
- Brand
- Inventory type
- Base unit
- Available quantity
- Reserved quantity
- On-order quantity
- Warehouse count
- Average cost
- Inventory value
- Reorder status
- Expiration warning
- Status
- Personal Star

11. Product Page Header

- Product image
- Name
- SKU
- Barcode/QR
- Product type
- Category
- Brand
- Base unit
- Current available quantity
- Inventory value
- Status
- Preferred supplier
- Last movement
- Last counted date

12. Product Page Actions

- Edit Product
- Receive Stock
- Issue Stock
- Transfer Stock
- Reserve Stock
- Adjust Stock
- Start Count
- Create Reorder Request
- Upload Document
- Add Supplier
- Archive

13. Product Identity and Master Data

- Product name
- SKU
- Alternate SKU
- Barcode
- QR code
- Manufacturer code
- Brand
- Category
- Subcategory
- Description
- Specifications
- Base unit
- Purchase unit
- Sales unit
- Weight
- Volume
- Dimensions
- Color
- Material
- Country of origin
- Tax category
- Status
- Tags

14. Units of Measure

- Base unit
- Purchase unit
- Issue unit
- Sales unit
- Conversion factor
- Rounding precision
- Minimum issue quantity
- Packaging unit
- Weight unit
- Volume unit
- Dimension unit
- Active/inactive status

INV-UOM-001 - Every stock movement is stored in the product base unit and retains the entered unit and conversion rate.

INV-UOM-002 - Unit conversions must not change historical movements when conversion settings are later edited.

15. Product Categories

- Hierarchical categories
- Category code
- Default unit
- Default valuation method
- Default tax code
- Default storage conditions
- Default reorder behavior
- Required tracking type
- Custom attributes
- Archive status

16. Brands and Manufacturers

- Brand name
- Manufacturer company
- Country
- Contact reference
- Warranty defaults
- Documentation
- Status

17. Custom Attributes

- Category-specific fields
- Text, number, date, select, multi-select, boolean and measurement field types
- Required/optional state
- Validation rules
- Searchable flag
- Filterable flag
- Display order
- Platform Configuration compatibility

18. Warehouse Model

- Company warehouse
- Project warehouse
- Site warehouse
- Transit warehouse
- Returns warehouse
- Quarantine warehouse

- Damaged-goods warehouse
- Virtual warehouse
- Consignment warehouse
- Custom warehouse type

19. Warehouse Page

- Warehouse identity
- Address and map reference
- Owning company
- Warehouse manager
- Operating hours
- Capacity
- Zones and bins
- Stock value
- Open receipts
- Open issues
- Pending transfers
- Low stock
- Count status
- Users
- Documents
- Tasks
- Activity

20. Warehouse Location Hierarchy

Recommended hierarchy: Warehouse -> Zone -> Aisle -> Rack -> Shelf -> Bin. Each level is optional so the same model supports simple and complex locations.

- Location code
- Parent location
- Capacity
- Allowed categories
- Temperature range
- Hazard classification
- Pick priority
- Put-away priority
- Status
- Barcode/QR

21. Stock Record

- Product
- Warehouse
- Location/bin
- Batch/lot
- Serial number
- Ownership type
- Quantity on hand
- Reserved quantity
- Available quantity
- Blocked quantity
- In-transit quantity
- Unit cost
- Inventory value
- Last movement
- Last count

Stock balances are derived from immutable movement records and may be cached for performance. They must

always be reproducible.

22. Stock Availability Formula

Available quantity = On-hand quantity - Reserved quantity - Blocked quantity. In-transit and on-order quantities are displayed separately and are not available until receipt.

23. Stock Statuses

- Available
- Reserved
- Blocked
- Quality Hold
- Quarantine
- Damaged
- Expired
- In Transit
- Consignment
- Pending Inspection
- Custom

24. Stock Movements

- Receipt
- Issue
- Transfer Out
- Transfer In
- Adjustment Increase
- Adjustment Decrease
- Reservation
- Reservation Release
- Return In
- Return Out
- Production Consumption
- Production Output
- Project Consumption
- Maintenance Consumption
- Count Adjustment
- Disposal

INV-MOV-001 - Every movement has a unique number, source, destination, date, quantity, unit, product, actor and status.

INV-MOV-002 - Posted stock movements are immutable. Corrections use reversing movements.

25. Movement Lifecycle

- Draft
- Under Review
- Approved
- Posted
- Partially Completed
- Completed
- Cancelled
- Reversed
- Archived

26. Purchase Receipts

- Purchase order reference

- Supplier
- Warehouse
- Receipt date
- Delivery note
- Products and quantities
- Expected vs received quantity
- Unit cost
- Batch/serial capture
- Quality status
- Damaged quantity
- Put-away location
- Documents
- Posting status

27. Goods Issue

- Requesting company/department/project
- Recipient user/team
- Warehouse
- Products
- Requested quantity
- Issued quantity
- Purpose
- Cost center
- Project
- Task/work order
- Approval reference
- Pick location
- Issue date
- Acknowledgement

28. Internal Transfers

- Origin warehouse/location
- Destination warehouse/location
- Transfer date
- Expected arrival
- Products
- Quantity
- Carrier/user
- In-transit status
- Partial receipt
- Variance
- Documents
- Approval
- Activity history

29. Returns

- Supplier returns
- Customer returns
- Project returns
- Unused material returns
- Damaged returns
- Return reason
- Condition
- Disposition
- Credit/debit note link
- Restock or quarantine decision

30. Reservations

- Product
- Quantity
- Warehouse
- Project
- Unit
- Client order
- Work order
- User/team
- Start date
- Expiry date
- Priority
- Status
- Partial release
- Automatic expiry later

INV-RES-001 - Reservations reduce available stock but do not reduce on-hand quantity.

31. Cycle Counts

- Count plan
- Warehouse/location scope
- Product/category scope
- Assigned counters
- Blind count option
- Expected quantity hidden option
- First count
- Recount
- Variance
- Approval
- Adjustment posting
- Count history

32. Physical Inventory

- Full warehouse count
- Freeze option
- Snapshot time
- Multiple counting teams
- Reconciliation
- Variance valuation
- Approval
- Adjustment journal link
- Final report

33. Batch and Lot Tracking

- Batch/lot number
- Manufacture date
- Receipt date
- Supplier
- Country of origin
- Expiration date
- Quantity
- Cost
- Quality status
- Documents
- Movement traceability
- Recall readiness

34. Serial Number Tracking

- Unique serial number
- Product
- Receipt
- Current warehouse/location
- Status
- Assigned user/project
- Warranty
- Movement history
- Conversion to Asset readiness

A serialized inventory item may later become an Asset without losing its procurement and stock history.

35. Expiration Management

- Expiration date
- Minimum remaining shelf life
- FEFO picking
- Expiring in 7/30/60/90 days
- Expired stock blocking
- Disposal workflow
- Supplier return option
- Notifications and reports

36. Quality and Quarantine

- Inspection required
- Quality hold
- Quarantine location
- Sample quantity
- Pass/fail
- Released quantity
- Rejected quantity
- Inspection document
- Responsible user
- Linked task/approval

37. Kits and Bundles

- Kit SKU
- Component products
- Component quantities
- Optional components
- Assembly/disassembly
- Availability calculation
- Bundle sales context
- Cost rollup
- Version history

38. Bill of Materials

- BOM code
- Parent product
- Version
- Components
- Quantities
- Units

- Waste factor
- Substitutes
- Effective dates
- Status
- Cost rollup
- Documents

BOM support is included for inventory planning. Full manufacturing execution is a future module.

39. Reorder Rules

- Minimum level
- Maximum level
- Reorder point
- Safety stock
- Lead time
- Order multiple
- Preferred supplier
- Warehouse-specific rule
- Seasonal rule
- Manual/automatic recommendation
- Status

40. Stock Replenishment

- Reorder recommendations
- Internal transfer recommendation
- Purchase request recommendation
- Shortage analysis
- Lead-time consideration
- Open order consideration
- Reserved demand consideration
- Approval readiness

41. Supplier Inventory Context

- Preferred supplier
- Supplier SKU
- Purchase unit
- Last purchase price
- Average lead time
- Minimum order quantity
- Price breaks
- Supplier availability manually recorded
- Supplier performance link

42. Inventory Valuation

- FIFO
- Weighted Average
- Specific Identification
- Standard Cost
- LIFO configurable but disabled where unsuitable
- Company/category/product valuation method
- Current inventory value
- Movement cost layers
- Valuation report

Finance owns accounting postings; Inventory calculates operational cost layers and sends posting-ready events.

43. Costing Rules

- Purchase cost
- Freight allocation
- Duty allocation
- Handling cost
- Landed cost
- Standard cost variance
- Write-down
- Obsolescence allowance
- Cost adjustment
- Currency conversion

44. Landed Cost

- Freight
- Customs
- Insurance
- Handling
- Broker fees
- Allocation by quantity, weight, volume or value
- Cost layer update
- Finance reference
- Audit history

45. Project Inventory

- Project warehouse
- Material reservation
- Material issue
- Return of unused stock
- Project consumption
- Building/floor/work-package allocation
- Budget comparison
- Cost-to-project
- Material shortage warnings

46. Asset and Maintenance Integration

- Spare-part consumption
- Consumables usage
- Serialized item conversion to asset
- Asset maintenance work order linkage
- Replacement part traceability
- Warranty and supplier link

47. Procurement Integration

- Purchase requisition source
- Purchase order expected receipts
- Partial receipt
- Over-receipt tolerance
- Supplier return
- Open-order quantity
- Landed cost context

- Three-way match readiness

48. Finance Integration

- Inventory asset account
- Cost of goods sold
- Expense accounts
- Goods received not invoiced
- Purchase price variance
- Stock adjustment
- Write-off
- Landed cost posting
- Project cost allocation
- Journal source references

49. Documents Integration

- Product specifications
- Certificates
- Safety data sheets
- Delivery notes
- Inspection records
- Warehouse procedures
- Count reports
- Photos
- Supplier documents
- Revisions through Document Passport

50. Tasks and Approvals Integration

- Restock task
- Count task
- Inspection task
- Transfer task
- Shortage resolution
- Adjustment approval
- Disposal approval
- High-value issue approval
- Participant-based task visibility

51. Search

- Product name
- SKU
- Barcode
- Serial number
- Batch/lot
- Category
- Brand
- Supplier SKU
- Warehouse
- Location
- Document text later
- Recent searches
- Pinned filters

52. Filters

- Company

- Warehouse
- Location
- Product type
- Category
- Brand
- Status
- Stock level
- Available/reserved
- Expiration
- Batch
- Serial tracked
- Supplier
- Project
- Last movement
- Last count
- Inventory value

53. Saved Views

- Private saved views
- Shared views later
- Column selection
- Sort order
- Filters
- Grouping
- Default view
- Pinned view
- Platform Configuration compatibility

54. Bulk Operations

- Bulk category assignment
- Bulk status change
- Bulk reorder rule
- Bulk supplier assignment
- Bulk location transfer
- Bulk export
- Bulk labels
- Bulk archive
- Bulk count scope

55. Import and Export

- Product master import
- Opening stock import
- Warehouse/location import
- Batch/serial import
- Validation preview
- Error report
- Duplicate handling
- Template download
- CSV/XLSX support
- Audit record

56. Barcode, QR and RFID Readiness

- Product labels
- Location labels
- Movement scanning

- Receiving scan
- Issue scan
- Transfer scan
- Count scan
- Serial scan
- Batch scan
- RFID integration future
- Mobile camera scanning

57. Inventory Reports

- Stock on hand
- Stock availability
- Inventory valuation
- Movement history
- Warehouse comparison
- Low stock
- Out of stock
- Overstock
- Dead stock
- Fast/slow moving
- Expiring stock
- Batch traceability
- Serial traceability
- Count variance
- Project consumption
- Supplier receipt performance
- Adjustments and write-offs

58. Analytics

- Inventory turnover
- Days inventory outstanding
- Stockout rate
- Fill rate
- Count accuracy
- Shrinkage
- Obsolescence
- Warehouse utilization
- Movement volume
- Reorder performance
- Project material usage
- Value by category

59. Inventory Settings

- Product types
- Categories
- Units of Measure
- Valuation methods
- Movement numbering
- Warehouse types
- Location hierarchy labels
- Stock statuses
- Tracking requirements
- Negative stock policy
- Reservation policy
- Count policy
- Expiration policy

- Default accounts
- Archive rules

60. Platform Configuration Compatibility

- Module toggle
- Page toggle
- Section toggle
- Widget toggle
- Feature toggle
- Workflow toggle
- Action toggle
- Report toggle
- Serial/batch toggle
- BOM toggle
- Valuation methods toggle
- Warehouse-specific override
- Company-specific override
- Group inheritance

Disabled functionality must disappear from navigation, search, APIs and layouts without empty spaces or dead links.

61. Responsive UX

Desktop

- Wide dashboard
- Sticky inventory navigation
- Dense but readable tables
- Split-view product/location panels
- Keyboard shortcuts

Tablet

- Two-column dashboard
- Collapsible navigation
- Horizontal movement tables
- Drawer-based detail

Mobile

- Scanner-first actions
- Card-based stock lists
- Bottom action sheet
- Compact movement forms
- Offline-ready architecture later
- Full core functionality

62. Premium Motion and Micro-Interactions

- Product card hover: 120-180 ms lift and border tint
- Stock quantity changes animate subtly
- Movement status transition
- Transfer progress indicator
- Count variance reveal
- Drawer slide 220-280 ms
- Skeleton loading matching final geometry
- Critical stock pulse only when necessary

- Reduced-motion compliance

63. Accessibility

- Keyboard navigation
- Visible focus states
- Accessible labels
- Color-independent status cues
- Screen-reader table headers
- Reduced motion
- Touch target minimums
- Readable contrast
- No hover-only actions

64. Empty, Loading and Error States

- No products
- No warehouses
- No stock
- No movements
- No count results
- No expiring items
- Module disabled
- Insufficient access later
- Source module unavailable
- Partial dashboard failure
- Archived product
- Import validation failure

65. Data Ownership Rules

- Inventory owns product stock configuration, warehouses, locations, movements, reservations, batches, serials and operational valuation layers.
- Procurement owns purchase requisitions, purchase orders and supplier selection.
- Finance owns accounting journals, payables and financial statements.
- Assets owns long-lived asset lifecycle after conversion.
- Projects owns project records and work context.
- Documents owns files and revisions.
- Tasks owns task execution and participant visibility.

66. Core Data Model

Entity	Purpose	Key fields
Product	Product master	id, tenant_id, company_id, sku, name, type_id, category_id, base_uom_id, tracking_type, status
ProductCategory	Hierarchy	id, parent_id, name, code, defaults_json, status
UnitOfMeasure	Unit and conversion	id, category, name, symbol, precision
ProductUOM	Product conversion	product_id, uom_id, factor, effective_from
Warehouse	Stock facility	id, company_id, type, code, name, address, manager_id, status
InventoryLocation	Location hierarchy	id, warehouse_id, parent_id, type, code, capacity, status
StockBalance	Cached balance	product_id, location_id, batch_id, serial_id, on_hand, reserved, blocked, value
InventoryMovement	Immutable movement	id, number, type, product_id, qty, uom_id, from_location_id, to_location_id, status, posted_at

MovementLine	Multi-line document	movement_id, product_id, qty, cost, batch_id, serial_id
Reservation	Stock reservation	id, product_id, warehouse_id, qty, source_type, source_id, expires_at, status
BatchLot	Batch tracking	id, product_id, lot_no, manufacture_at, expires_at, supplier_id, status
SerialNumber	Serial tracking	id, product_id, serial_no, status, current_location_id
InventoryCount	Count header	id, warehouse_id, scope_json, snapshot_at, status
InventoryCountLine	Count result	count_id, product_id, location_id, expected_qty, counted_qty, variance
ReorderRule	Replenishment rule	product_id, warehouse_id, min_qty, max_qty, reorder_point, lead_days
BOM	Bill of materials	id, parent_product_id, version, status
BOMLine	BOM component	bom_id, component_product_id, qty, uom_id, waste_pct
InventoryCostLayer	Costing layer	product_id, movement_id, qty_remaining, unit_cost, currency
InventoryActivity	Audit/activity	actor_id, entity_type, entity_id, event_type, occurred_at

67. Database Constraints and Indexing

- Unique SKU within configured company/group scope
- Unique serial number within tenant
- Warehouse and location hierarchy cycle prevention
- Posted movement append-only protection
- Indexes on product, warehouse, location, date, batch, serial, source and status
- Partition movement tables by tenant/company/date when needed
- Decimal quantities and costs; never floating-point
- Optimistic concurrency for master data

68. API Requirements

- GET /inventory/overview
- GET/POST /products
- GET/PATCH /products/{id}
- POST /products/import
- GET/POST /warehouses
- GET/POST /warehouses/{id}/locations
- GET /stock
- GET /stock/availability
- GET/POST /inventory-movements
- POST /inventory-movements/{id}/approve
- POST /inventory-movements/{id}/post
- POST /inventory-movements/{id}/reverse
- POST /receipts
- POST /issues
- POST /transfers
- POST /returns
- GET/POST /reservations
- POST /reservations/{id}/release
- GET/POST /inventory-counts
- POST /inventory-counts/{id}/complete
- GET /batches
- GET /serial-numbers
- GET/POST /reorder-rules

- GET/POST /boms
- GET /inventory/valuation
- GET /inventory/reports

Posting endpoints must be idempotent and require idempotency keys to prevent duplicate stock movements.

69. API Response Principles

- Cursor pagination
- Consistent filters
- Tenant/company context required
- Platform Configuration enforced
- Source record links
- Decimal values as strings
- Version fields for concurrency
- Structured validation errors
- Audit correlation ID

70. Frontend Architecture

- InventoryDashboardPage
- ProductsPage
- ProductPage
- InventoryFilterBar
- ProductCard
- StockTable
- WarehousesPage
- WarehousePage
- LocationTree
- MovementWorkspace
- ReceiptForm
- IssueForm
- TransferForm
- ReservationPanel
- CountWorkspace
- BatchPanel
- SerialPanel
- ReorderPanel
- BOMEditor
- ValuationPage
- InventoryReportsPage
- InventorySettingsPage
- ScannerDrawer

71. Frontend State Management

- URL-based filter state
- Server-state cache
- Lazy loading by section
- Optimistic low-risk master-data edits
- No optimistic stock posting
- Mutation idempotency
- Virtualized large tables
- Preserved scanner session
- Draft form recovery
- Real-time/event refresh readiness

72. Backend Services

- Product master service
- Warehouse/location service
- Stock ledger service
- Movement posting engine
- Reservation service
- Batch/serial service
- Count service
- Replenishment service
- BOM service
- Valuation service
- Import/export service
- Search indexing service
- Activity/audit service
- Configuration resolver
- Cache invalidation service

73. Domain Events

- product.created
- stock.received
- stock.issued
- stock.transferred
- stock.reserved
- stock.released
- stock.adjusted
- stock.counted
- stock.expiring
- stock.low
- batch.quarantined
- serial.converted_to_asset
- inventory.value_changed

74. Background Jobs

- Expiration status calculation
- Low-stock evaluation
- Reservation expiry later
- Reorder recommendations
- Inventory valuation snapshot
- Dead-stock classification
- Search indexing
- Import processing
- Report generation
- Count reminders
- Cache reconciliation

All background jobs must be idempotent, tenant-aware and safe to retry.

75. Validation Rules

- Product name and SKU required
- Valid unit conversions
- No negative quantity unless explicitly enabled
- Serial quantity must equal whole units
- Batch required for batch-tracked products
- Expiration required when category demands it

- Origin/destination cannot be identical
- Destination warehouse/location active
- Available stock check before issue
- Posted movement cannot be edited
- Count variance adjustment requires approval when configured
- Reservation cannot exceed configured availability policy
- Closed period finance impact handled safely

76. Security and Permission Readiness

- Tenant isolation
- Company and warehouse scope
- High-value adjustment permission readiness
- Posting/reversal permission readiness
- Cost visibility readiness
- Supplier price confidentiality readiness
- Audit of sensitive changes
- No linked-module access expansion
- Section/field/action permission readiness
- Platform Configuration evaluated separately

77. Audit and Immutability

- Every movement stores actor, date, source, before/after context and correlation ID
- Posted movements immutable
- Reversal linked to original
- Master-data changes logged
- Count results preserved
- Import and bulk actions logged
- Valuation changes traceable
- Audit events immutable

78. Performance and Scalability

- Append-only movement ledger
- Cached balances rebuilt from movements
- Cursor pagination
- Virtualized tables
- Batch API requests
- Database partitioning readiness
- Warehouse/product/date indexes
- Asynchronous heavy reports
- Support millions of movement lines
- Dashboard target under 2 seconds
- Scanner response target under 300 ms

79. Analytics and Telemetry

- Page and feature usage
- Movement processing time
- Import error rate
- Scanner success rate
- Dashboard failures
- Count completion time
- Stockout events
- Reorder recommendation acceptance
- API latency
- No employee-surveillance analytics

80. Testing Strategy

Unit Tests

- Unit conversion
- Availability formula
- Movement balance
- Serial/batch validation
- Reservation logic
- Count variance
- FIFO/average costing
- Reorder calculation
- BOM cost rollup
- Platform Configuration resolution

Integration Tests

- Procurement receipt
- Finance posting event
- Project issue
- Asset spare-part usage
- Documents
- Tasks/approvals
- Import/export
- Activity events

End-to-End Tests

- Create product
- Create warehouse/location
- Receive stock
- Reserve stock
- Issue stock
- Transfer and receive
- Batch/serial capture
- Physical count
- Adjustment approval
- Generate valuation
- Disable feature through Platform Configuration
- Mobile scanning

Visual Regression

- Dashboard desktop
- Products desktop
- Warehouse page
- Count workspace
- Tablet
- Mobile
- Long SKU/product name
- Disabled-section reflow
- Reduced-motion mode

81. Acceptance Criteria

- AC-01 - Inventory supports multiple companies and warehouses.
- AC-02 - Products support configurable categories, types and units.
- AC-03 - Stock balances are reproducible from immutable movements.
- AC-04 - Receiving, issuing, transfers, returns and adjustments are implemented.
- AC-05 - Reservations reduce availability correctly.

- AC-06 - Cycle counts and physical inventory support variance approval.
- AC-07 - Batch, serial and expiration tracking are supported.
- AC-08 - Reorder rules and recommendations are available.
- AC-09 - Kits and BOM structures are supported.
- AC-10 - FIFO, weighted average and specific identification are supported.
- AC-11 - Project, Assets, Procurement and Finance integrations are source-linked.
- AC-12 - Barcode/QR scanning workflows are supported.
- AC-13 - Desktop, tablet and mobile retain core functions.
- AC-14 - Premium motion is implemented.
- AC-15 - Reduced-motion preference is respected.
- AC-16 - Every inventory element supports Platform Configuration.
- AC-17 - Disabled features leave no dead links or blank spaces.
- AC-18 - Audit history is immutable.
- AC-19 - Large movement volumes remain performant.
- AC-20 - The module is ready for future field-level permissions.

82. Page and Flow Registry

- Inventory Dashboard
- Products Module
- Product Page
- Create Product
- Edit Product
- Archive Product
- Warehouses
- Warehouse Page
- Create Warehouse
- Location Setup
- Receive Stock
- Issue Stock
- Transfer Stock
- Receive Transfer
- Return Stock
- Reserve Stock
- Release Reservation
- Stock Adjustment
- Cycle Count
- Physical Inventory
- Batch Page
- Serial Number Page
- Expiration Management
- Quality Hold
- Kits and Bundles
- BOM Editor
- Reorder Rules
- Replenishment Recommendations
- Inventory Valuation
- Inventory Reports
- Inventory Import
- Label Printing
- Inventory Settings

83. Implementation Phases

1. Foundation: Product master, categories, units, warehouses, locations and Platform Configuration.
2. Stock ledger: receipts, issues, transfers, returns, adjustments and balances.
3. Control: reservations, counts, serials, batches, expiration and quality hold.
4. Planning: reorder rules, supplier context, kits, BOM and replenishment.
5. Finance and analytics: valuation, costing, project/asset/procurement integrations and reports.

6. Hardening: performance, accessibility, audit, security and complete testing.

84. Future Roadmap

- RFID reader integrations
- IoT warehouse sensors
- Courier integrations
- EDI supplier documents
- Automated bank/ERP integration
- Advanced manufacturing execution
- Wave picking
- Route optimization
- Voice picking
- AI demand forecasting
- AI anomaly and shrinkage detection
- Offline mobile synchronization

Future integrations and AI capabilities are not part of the initial implementation.

85. Final Architectural Rule

Inventory owns stock identity, quantity, location, movement and operational valuation. Procurement owns purchasing, Finance owns accounting, Assets owns long-lived asset lifecycle, Projects owns work context, and the Inventory Module links these records without duplicating ownership.